CLEF 2026
BioASQ Lab · Task 14b
Jena, Germany · 21–24 September 2026

# Agentic RAG with Open-Weight LLMs for Biomedical QA (BioASQ Task 14b)

Rohan Leekha, Daniel Gwon, and Leslie Shing – MIT Lincoln Laboratory

## SUMMARY / KEY FEATURES

- Two open-weight agentic RAG systems submitted to all four BioASQ Task 14b batches: ossllm (Gemma 3 27B + live NCBI E-utilities) and Finalcorrected (Gemma 4 31B Dense + local SQLite FTS5 mirror of PubMed) [6]
- A single LLM plays both controller and generator: it issues PubMed queries, inspects the evidence pool, and chooses search / sufficient / stop under a hard cap of N = 5 iterations
- Hybrid retrieval: PubMedBERT dense embeddings + BM25 sparse ranking fused with Reciprocal Rank Fusion (k = 60) [3]
- Type-specific prompts for yes/no, factoid, list, and summary questions plus a deterministic format-hygiene filter for BioASQ's strict scorer
- No domain fine-tuning: 0.941 yes/no accuracy on Phase B batch 1 (upper-middle quartile of the field); Phase A+ results bounded by retrieval recall

## SUBMITTED SYSTEMS

| | ossllm | Finalcorrected |
|---|---|---|
| **LLM** | Gemma 3 27B IT | Gemma 4 31B Dense |
| **PubMed access** | live NCBI E-utilities | local SQLite FTS5 mirror |
| **Re-ranking** | PMB + BM25 + RRF | PMB + FTS5 + RRF |
| **Few-shot** | 3 training exemplars | 3 training exemplars |
| **Batches (A+ / B)** | 2, 3, 4 / 1 | 4 / 3, 4 |

**Contact: rohan.leekha@ll.mit.edu**
Source code available on request from the authors · CLEF 2026 Working Notes

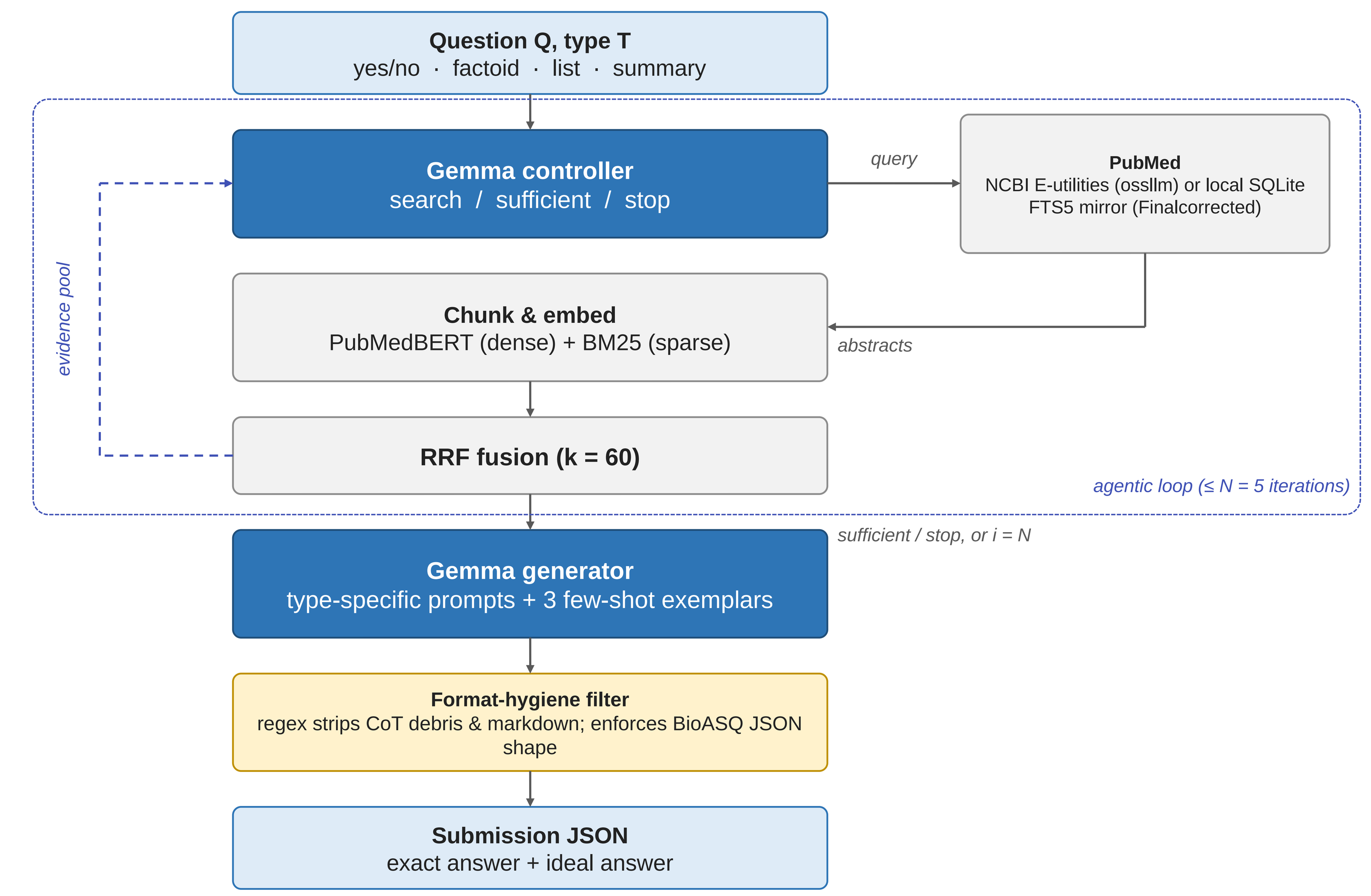
## MOTIVATION

BioASQ Task 14b [1] asks systems to retrieve PubMed evidence (Phase A), answer yes/no, factoid, list, and summary questions from gold snippets (Phase B), and do both end-to-end (Phase A+), where retrieval errors propagate straight into the answer. We test how far a stripped-down, single-model, open-weight agentic RAG system can go with no domain fine-tuning — and which component of the pipeline is the binding constraint.

## PROBLEM

- On Phase A+ every retrieval miss is an answer miss: a gold entity that never enters the evidence pool cannot be listed
- Instruction-tuned LLMs are sycophantic — the base Gemma 3 prompt answered "yes" to ~70% of training yes/no questions regardless of evidence polarity [5]
- BioASQ's exact-match scorer rejects full-sentence factoids, under-recalled lists, and chain-of-thought debris that leaks into the JSON
- A naive agentic controller paraphrase-cycles one query and talks itself into stopping before the evidence pool is complete
- Learned rerankers and fine-tuned generators dominate the 13b leaderboard [2]; we omit them deliberately to isolate what an agentic loop plus format hygiene can do alone

## APPROACH




A single Gemma instance is both controller (choose the next PubMed query, or declare the evidence pool sufficient) and generator, in the spirit of ReAct [4]. PubMed is a data tool only: ranking, RRF fusion, deduplication, and chunk selection run client-side, so the LLM steers retrieval purely through its query reformulations. The loop ends on sufficient / stop or after N = 5 iterations; termination triggers type-specific generation and the deterministic format-hygiene pass.

## KEY FINDINGS

- **0.941 yes/no accuracy** on Phase B batch 1 with no domain fine-tuning — upper-middle quartile of the field (leader 1.000; ~20 systems tied at 0.941)
- **Retrieval recall binds end-to-end quality:** on Phase A+ batch 4, swapping Gemma 3 27B → Gemma 4 31B Dense and live E-utilities → a local PubMed mirror left all five metrics unchanged to three decimals
- **Format hygiene pays:** the deterministic filter recovered ~20–30 list answers that the strict scorer would otherwise have rejected as malformed

| Metric (best batch) | Ours | Leader | Position |
|---|---|---|---|
| Phase B yes/no accuracy (b1) | **0.941** | 1.000 | Upper-middle quartile |
| Phase B list F-measure (b3) | 0.365 | ~0.50 | Middle of field |
| Phase B factoid strict (b3) | 0.364 | ~0.53 | Middle of field |
| Phase B ideal-answer ROUGE-2 F1 (b1) | 0.179 | ~0.25 | Middle of field |
| Phase A+ yes/no accuracy (b4) | 0.750 | 0.938 | Lower-middle |
| Phase A+ list F-measure (b4) | 0.255 | ~0.58 | Lower-middle |

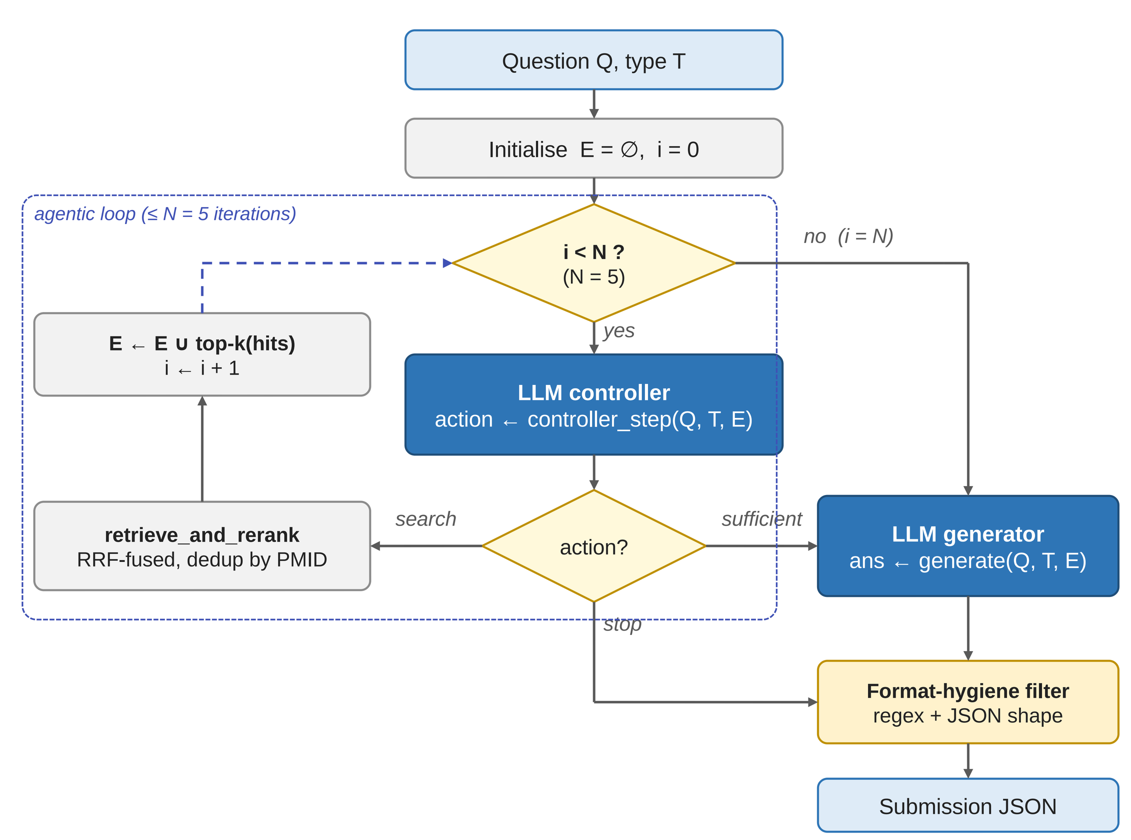
*Field-leader numbers are approximate, read from the public leaderboard [2] at the time of writing.*

## FUTURE WORK

- Learned cross-encoder reranker on BioASQ relevance judgments — the single highest-leverage change for Phase A+ recall
- MeSH term expansion and biomedical entity normalization on outgoing queries
- Per-type prompt re-tuning with explicit word caps when swapping backbones (closes the ROUGE-2 gap)
- Controlled ablations of the iteration cap, sufficiency check, and format-hygiene filter; generation-time ensembles

## AGENTIC CONTROL LOOP & TYPE-SPECIFIC GENERATION

- **What the controller sees:** the question, its type, and the evidence pool — but not its own query history. It must ask "does what I have answer this?" rather than "have I searched enough?", which also defuses paraphrase-cycling (NF1 glioblastoma → glioblastoma NF1 loss → neurofibromin in GBM …)
- **How it decides to stop:** before choosing search or sufficient it writes a hidden checklist of the claims the answer must support and checks the pool against each one — compound questions earn extra queries, simple lookups exit after one round




### Sufficiency checklist in action

**Q (factoid):** *"In GBM, which cell state is associated with loss of NF1?"*

**Iteration 1 · query** *NF1 loss glioblastoma* → pool covers NF1's role in GBM, not the cell-state taxonomy

**Checklist:** 1) GBM cell-state taxonomy ✗ unsupported 2) state tied to NF1 loss ✓ supported → **search**

**Iteration 2 · query** *glioblastoma cell-state taxonomy* → both claims supported → **sufficient**

**Generator (factoid prompt) → exact answer: mesenchymal**

A simple lookup ("What is the indication for Brineura?") satisfies every claim on the first round and exits after one iteration — search budget goes to the questions that are genuinely compound.

### Type-specific prompts + format-hygiene filter

- **Yes/no:** enumerate evidence against, then for, then commit — counters the ~70% "yes" bias; the prompt states "no" is right about half the time
- **Factoid:** entity only; post-processor truncates to the first noun phrase (1–5 tokens), up to 3 ranked candidates
- **List:** exhaustive extraction over every passage, then normalized case-insensitive dedup
- **Summary:** prefer named entities and numbers over generic phrasing; no exact_answer field
- **Filter:** regex strips CoT debris and markdown, unwraps prefixes, and enforces the BioASQ ≥5 JSON shape (≤5 factoid, ≤100 list entries)

## RESULTS

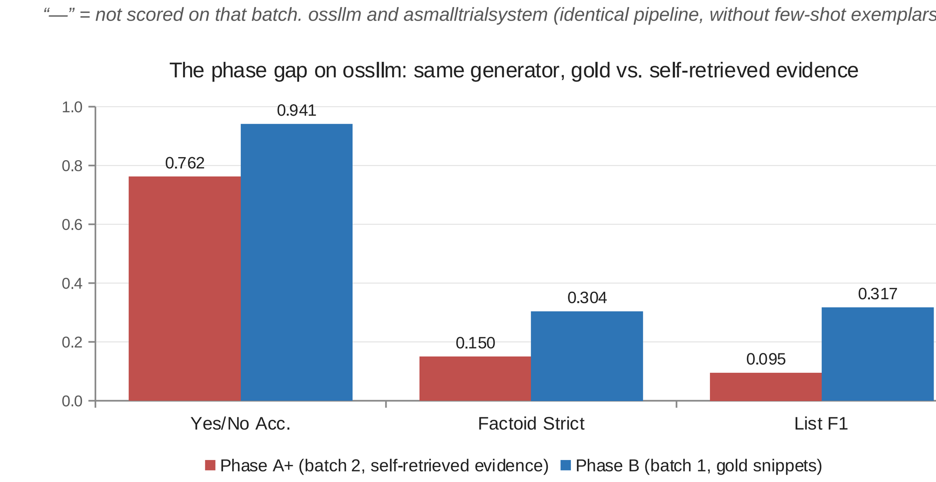
**Phase B — answering from gold snippets (exact answers + ROUGE-2 F1 on ideal answers)**

| Batch | System | Y/N Acc. | Y/N Macro-F1 | Factoid Strict | Factoid MRR | List F1 | Ideal R-2 F1 |
|---|---|---|---|---|---|---|---|
| 1 | ossllm | **0.941** | 0.938 | 0.304 | 0.370 | 0.317 | 0.179 |
| 3 | Finalcorrected | 0.813 | 0.806 | 0.364 | 0.364 | 0.365 | 0.158 |
| 4 | Finalcorrected | 0.813 | 0.806 | 0.364 | 0.364 | 0.365 | 0.087 |

**Phase A+ — end-to-end: retrieve, then answer**

| Batch | System | Y/N Acc. | Y/N Macro-F1 | Factoid Strict | Factoid MRR | List F1 |
|---|---|---|---|---|---|---|
| 2 | ossllm | 0.762 | 0.760 | 0.150 | 0.150 | 0.095 |
| 3 | ossllm | 0.455 | 0.450 | — | — | 0.201 |
| 4 | ossllm | 0.750 | 0.746 | 0.273 | 0.273 | 0.255 |
| 4 | Finalcorrected | 0.750 | 0.746 | 0.273 | 0.273 | 0.255 |

*"—" = not scored on that batch. ossllm and asmalltrialsystem (identical pipeline, without few-shot exemplars) returned identical scores on every shared batch.*




- Same generator, gold snippets vs. self-retrieved evidence: the gap grows from yes/no → factoid → list, ordered by how recall-sensitive the metric is
- Finalcorrected on batch 4: 0.813 / 0.364 / 0.365 (Phase B) vs. 0.750 / 0.273 / 0.255 (Phase A+) — a 6-, 9-, and 11-point gap
- Batch-to-batch ROUGE-2 swings (0.158 → 0.087) come from verbosity calibration, not retrieval: Gemma 4 writes longer summaries than Gemma 3

[1] A. Nentidis, G. Katsimpras, A. Krithara, G. Paliouras. BioASQ at CLEF 2026: the fourteenth edition of the large-scale biomedical semantic indexing and question answering challenge. ECIR 2026.
[2] BioASQ. Large-scale biomedical semantic indexing and question answering, https://www.bioasq.org/ (leaderboard accessed 2026-05-27); R. A. A. Jonker et al. BIT.UA at BioASQ 13b. CLEF 2025 Working Notes.
[3] G. V. Cormack, C. L. A. Clarke, S. Buettcher. Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. SIGIR 2009.
[4] S. Yao et al. ReAct: synergizing reasoning and acting in language models. ICLR 2023.
[5] M. Sharma, M. Tong, T. Korbak et al. Towards understanding sycophancy in language models. ICLR 2024.
[6] Gemma Team. Gemma 3 technical report. arXiv:2503.19786 (2025); D. Chou, C. Chang. Gemma 4: byte for byte, the most capable open models. Google, 2026.

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited. This material is based upon work supported by the Department of the Air Force under Air Force Contract No. FA8702-15-D-0001 or FA8702-25-D-B002. Any opinions, findings, conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the Department of the Air Force. 

LINCOLN LABORATORY
MASSACHUSETTS INSTITUTE OF TECHNOLOGY